DAY 00 / COMPLETE MASTER INDEX

Cloud and LLMOps Learning Index

A complete map of Days 1-24, from retrieval and agentic systems through engineering, models, cloud infrastructure, delivery, and LLM operations.

25	903	24
TOPIC GUIDES	PDF PAGES	LEARNING DAYS

RECOMMENDED PATH / 6 PHASES

1 — DAYS 1-3

Retrieval foundations

Ground answers in trusted enterprise data.

2 — DAYS 4-7

Orchestration and integration

Control workflows and connect external systems.

3 — DAYS 8-11

Engineering foundations

Strengthen design, coding, and architecture skills.

4 — DAYS 12-17

Model and inference foundations

Understand models before choosing infrastructure.

5 — DAYS 18-20

Containers, infrastructure, and runtime

Package, provision, and run the platform.

6 — DAYS 21-24

Delivery, GitOps, and LLM operations

Automate change and measure AI behavior.

HOW TO USE THIS INDEX

Follow the phases in order for a complete learning path, or use the quick locator to jump directly to a capability and its guide.

PHASE 01 / DAYS 1-3

Retrieval foundations

Build the core retrieval pattern, then apply data-focused and application-focused frameworks.

DAY 1

Vanilla RAG End to End

Builds the baseline retrieval pipeline from document ingestion through grounded answer generation.

KEY COVERAGE RAG architecture; loading; chunking; embeddings; vector search; retrieval; prompt construction; citations; evaluation; production safeguards

PDF 51 pages

DAY 2

LlamaIndex Foundations and Workflows

Uses a data-focused framework to ingest, index, retrieve, cite, evaluate, and query enterprise information.

KEY COVERAGE documents and nodes; readers; indexes; retrievers; query and chat engines; metadata; persistence; evaluation; workflows

PDF 34 pages

DAY 3

LangChain End to End

Assembles production LLM applications from reusable model, prompt, retrieval, tool, and output components.

KEY COVERAGE model interfaces; prompt templates; chains; retrievers; tools; structured output; agents; callbacks; testing; observability

PDF 27 pages

MEMORY LINE

RAG is the grounding pattern. LlamaIndex organizes enterprise data. LangChain assembles the application around models, prompts, retrieval, and tools.

PHASE 02 / DAYS 4-7

Orchestration and integration

Add stateful execution, standardized connectivity, architecture boundaries, and an integrated production view.

DAY 4

LangGraph AI Workflows

Designs controlled, stateful AI workflows with explicit decisions, recovery, approval, and persistence.

KEY COVERAGE state, nodes, and edges; branching; checkpointing; retries; memory; human-in-the-loop; streaming; durable execution

PDF 33 pages

DAY 5

MCP End-to-End Overview

Explains the standard protocol used by AI applications to discover and access external tools and data.

KEY COVERAGE hosts; clients; servers; tools; resources; prompts; discovery; transports; permissions; security; governance; versioning

PDF 38 pages

DAY 6

How RAG, LlamaIndex, LangChain, LangGraph, and MCP Fit Together

Maps the five technologies to distinct architectural roles and shows when they should be combined.

KEY COVERAGE category boundaries; retrieval versus tools; data versus live state; orchestration; integration patterns; decision framework

PDF 40 pages

DAY 7

The Complete Production AI Stack - Revision

Reviews the end-to-end stack and connects retrieval, workflows, tools, APIs, safety, testing, and operations.

KEY COVERAGE architecture layers; retrieval path; tool path; workflow control; security; evaluation; observability; deployment trade-offs

PDF 51 pages

MEMORY LINE

LangGraph controls the workflow. MCP standardizes external connectivity. The integrated stack assigns every layer a clear responsibility.

PHASE 03 / DAYS 8-11

Engineering foundations

Prepare for senior engineering decisions with system design, reusable DSA patterns, and clean GenAI architecture.

DAY 8

Lead Engineer System Design and LLD Playbook

Provides a repeatable interview method for turning ambiguous requirements into production-ready architectures.

KEY COVERAGE requirements; capacity estimates; APIs; data models; caching; queues; consistency; resilience; security; AI workloads; LLD

PDF 40 pages

DAY 9

DSA Core I - Arrays, Strings, Hashing, and Prefix Sums

Builds recognition of high-value array and string patterns for efficient interview solutions.

KEY COVERAGE Big-O; traversal; sets and dictionaries; frequency maps; complement lookup; prefix sums; Python complexity; edge cases

PDF 30 pages

DAY 10

DSA Core II - Two Pointers, Sliding Window, Stacks, and Queues

Replaces nested-loop thinking with efficient one-pass patterns for ranges, order, and stream processing.

KEY COVERAGE opposite and same-direction pointers; fixed and variable windows; stacks; queues; deques; pattern recognition; complexity

PDF 31 pages

DAY 11

Design Patterns and Clean Architecture for GenAI Systems

Separates business logic from providers, databases, model APIs, and framework-specific implementation details.

KEY COVERAGE layers; dependency inversion; ports and adapters; Strategy; Factory; Adapter; Facade; Repository; testing; provider abstraction

PDF 50 pages

MEMORY LINE

System design explains the whole service. DSA patterns sharpen implementation. Clean architecture keeps GenAI components replaceable and testable.

PHASE 04 / DAYS 12-17

Model and inference foundations

Move from classical machine learning to transformers, LLM behavior, multimodality, adaptation, and production inference.

DAY 12

ML Fundamentals and Interview Preparation

Builds a practical machine-learning foundation across learning types, evaluation, classical models, NLP, and vision.

KEY COVERAGE supervised, unsupervised, and reinforcement learning; splits; leakage; regularization; metrics; classical algorithms

PDF 21 pages

DAY 13

Neural Networks and Transformers

Explains deep-learning mechanics before deriving the transformer architecture and attention mental model.

KEY COVERAGE layers; activations; loss; backpropagation; SGD and Adam; self-attention; Q, K, and V; multi-head attention; residuals

PDF 23 pages

DAY 14

LLM Fundamentals and Model Families

Connects tokenization, training stages, inference behavior, sampling, context limits, and model-family trade-offs.

KEY COVERAGE BPE and SentencePiece; pre-training; instruction tuning; RLHF and DPO; temperature; top-k and top-p; context windows

PDF 21 pages

DAY 15

Multimodal Models and Diffusion

Introduces systems that combine language with vision and explains diffusion-based generation and its limitations.

KEY COVERAGE vision encoders; adapters; projection layers; cross-attention; multimodal use cases; OCR limits; diffusion training and generation

PDF 30 pages

DAY 16

Training, Fine-Tuning, PEFT and Evaluation

Provides a decision framework for adapting models, preparing datasets, and evaluating behavior efficiently.

KEY COVERAGE full fine-tuning; LoRA and QLoRA; continued pre-training; SFT; preference tuning; data quality; PII removal; evaluation

PDF 28 pages

DAY 17

LLM Inference, Deployment and Monitoring

Moves models into production through hosted or self-hosted inference, optimization, deployment, and controls.

KEY COVERAGE API versus self-hosted; vLLM, TGI, and Ollama; TTFT; throughput; batching; caching; quantization; streaming; rollback

PDF 26 pages

MEMORY LINE Data and training shape model capability. Inference architecture shapes latency, cost, reliability, and the operational experience.

PHASE 05 / DAYS 18-20

Containers, infrastructure, and runtime

Package the application, declare its AWS foundation, and operate services through Kubernetes and Helm.

DAY 18A

Docker and Docker Compose for GenAI

Packages a production-style GenAI application and its dependencies into reliable, observable local services.

KEY COVERAGE images and containers; layers; ports; environment variables; volumes; networks; Dockerfiles; health checks; Compose

PDF 55 pages

DAY 18B

Terraform and IaC Fundamentals for AWS

Introduces declarative infrastructure, Terraform language and workflow, state, and reusable AWS foundations.

KEY COVERAGE providers and resources; variables; locals; outputs; init, plan, and apply; state; S3 backends; locking; modules

PDF 33 pages

DAY 19

AWS GenAI Infrastructure with Terraform

Designs the cloud foundation for a production GenAI SaaS platform with managed AWS services and Terraform.

KEY COVERAGE VPCs and subnets; NAT and endpoints; security groups; EKS; ALB; RDS PostgreSQL; Redis; S3; ECR; IAM; autoscaling

PDF 30 pages

DAY 20

Kubernetes and Helm for GenAI Applications

Deploys FastAPI and React services through Kubernetes resources and reusable Helm charts.

KEY COVERAGE Pods; Deployments; Services; Ingress; ConfigMaps; Secrets; chart structure; values; templates; probes; scaling; rollouts

PDF 30 pages

MEMORY LINE Docker packages the workload. Terraform provisions the cloud foundation. Kubernetes runs and scales services. Helm makes releases reusable.

PHASE 06 / DAYS 21-24

Delivery, GitOps, and LLM operations

Automate testing and release, reconcile desired state, measure AI quality, and integrate the enterprise platform.

DAY 21

Jenkins CI/CD for GenAI

Builds a delivery pipeline that tests, packages, publishes, and deploys a GenAI application to EKS.

KEY COVERAGE controllers and agents; pipelines; lint and tests; RAG golden tests; Docker builds; ECR; Helm; credentials; rollback

PDF 27 pages

DAY 22

Argo CD GitOps for GenAI Platforms

Reframes deployment as pull-based reconciliation with Git as the auditable source of truth.

KEY COVERAGE desired state; drift detection; CI/CD boundaries; Argo CD components; repository design; sync; promotion; self-healing

PDF 55 pages

DAY 23

MLflow, MLOps and LLMOps

Tracks experiments and production behavior across prompts, traces, evaluations, datasets, feedback, and promotion.

KEY COVERAGE tracking contracts; parameters; metrics; artifacts; prompt registry; GenAI tracing; scorers; datasets; feedback; model registry

PDF 46 pages

DAY 24

Environments and Final End-to-End Integration

Combines the full platform into an operating model with explicit contracts, ownership, isolation, and lifecycle boundaries.

KEY COVERAGE target architecture; cross-system contracts; ownership; environment isolation; configuration; secrets; request flow; re-indexing

PDF 53 pages

MEMORY LINE

Jenkins proves and packages change. GitOps reconciles desired state. MLflow measures behavior. Environment contracts keep the platform governable.

QUICK LOCATOR / DAYS 1-13

Find the right guide fast

Use this cross-reference when you know the capability you need but not the day that covers it.

Need to find	Start with	Guide	Pages
Ground answers in private documents	Vanilla RAG	**Day 1**	51
Ingest, index, and query enterprise data	LlamaIndex	**Day 2**	34
Connect models, prompts, retrieval, and tools	LangChain	**Day 3**	27
Branching, state, retries, and approvals	LangGraph	**Day 4**	33
Standard tool and data connectivity	MCP	**Day 5**	38
Choose and combine the AI frameworks	Integration map	**Day 6**	40
Review the complete production AI stack	Full-stack revision	**Day 7**	51
Requirements, scale, architecture, and LLD	System design	**Day 8**	40
Arrays, hashing, and prefix sums	DSA Core I	**Day 9**	30
Pointers, windows, stacks, and queues	DSA Core II	**Day 10**	31
Maintainable provider-neutral GenAI code	Clean architecture	**Day 11**	50
Metrics, leakage, and classical ML	ML fundamentals	**Day 12**	21
Attention and transformer architecture	Deep learning	**Day 13**	23

PATH NOTE Days 1-11 establish application patterns and engineering depth. Continue with Days 12-24 for models, infrastructure, delivery, and operations.

QUICK LOCATOR / DAYS 14-24

Continue through the platform stack

Locate model, infrastructure, delivery, and operations topics across the second half of the collection.

Need to find	Start with	Guide	Pages
Tokens, sampling, context, and models	LLM behavior	**Day 14**	21
Vision-language systems and diffusion	Multimodal generation	**Day 15**	30
LoRA, QLoRA, SFT, and evaluation	Model adaptation	**Day 16**	28
Serving, latency, caching, and rollback	Inference deployment	**Day 17**	26
Containers and a local service stack	Docker and Compose	**Day 18A**	55
Terraform language, workflow, and state	Infrastructure as Code	**Day 18B**	33
VPC, EKS, RDS, Redis, S3, and IAM	AWS foundation	**Day 19**	30
Pods, Deployments, and Helm charts	Kubernetes runtime	**Day 20**	30
Build, test, publish, and deploy	Jenkins CI/CD	**Day 21**	27
Drift, reconciliation, and promotion	Argo CD GitOps	**Day 22**	55
Prompts, traces, evaluation, and registry	MLflow and LLMOps	**Day 23**	46
Environments, contracts, and ownership	Final integration	**Day 24**	53

COLLECTION SUMMARY 25 guides across 24 learning days. 903 total PDF pages. Six phases connect retrieval, orchestration, engineering, models, cloud, delivery, and operations.

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